String matching exercises

32-E01 Overlap, empty patterns, and comparisons

Use naive matching on `T=aaaaa`, `P=aaa`. List all shifts and character comparisons. Apply this chapter's specification to the empty pattern and `P=aaaaaa`. Explain why the algorithm cannot read `P[1]` before handling these cases.

32-E02 A collision is not a match

For `T=152615`, `P=15`, $d = 10, q = 11$, list every rolling hash and verification result starting from the first window. Derive each new hash from its predecessor rather than only reducing each window directly. Which shifts are falsely accepted if character verification is omitted?

32-E03 All borders and prefix construction

Build the full prefix function for `P=abacababac`. List every proper border of P_{10} and prove the failure chain skips none. What goes wrong if $\pi[q] = q$?

32-E04 Accepting states also transition

For `P=aba` and $\Sigma = \{a, b\}$, build the full transition table for states 0 through 3. Scan `T=abababa`, listing every state and output. Give KMP's retained state after each match. Explain when its state equals the automaton state and when it differs.

32-E05 Many comparisons for one character

Let $P = a^k b, T = a^{2k} c, k \geq 2$. Trace the first $k + 1$ text characters and fallback on the final `c`. Prove total fallback count is $O(n)$ rather than claiming worst-case $O(1)$ per character.

32-E06 Conditions for a random base

For fixed distinct equal-length strings X, Y, choose a uniform base in prime field $\mathbb{F}_q$. Prove collision probability at most $(m - 1)/q$ when differing character codes remain distinct modulo q. If $q = 5$ and character codes 0 and 5 are both allowed, construct a counterexample and identify the missing premise. Can total expected spurious hits still be bounded when window events are dependent?

String matching exercises (continued)

32-E07 Streaming across chunks

The text arrives as chunks `aabaa` and `abaaab`, with pattern `aabaaab`. Design a KMP state that stores no old text. List matches crossing chunk boundaries. Prove that the prefix function, current q, and total characters read are sufficient. Use global shifts rather than chunk-local indices.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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